Machine Learning and Traditional Econometrics for Volatility Forecasting:

Statistical Accuracy and Portfolio Implications

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ABSTRACT

Stock markets are built around the fundamental trade-off that higher expected returns on investment come at the cost of greater expected risk, as measured by volatility. As financial markets are forward-looking, they are constantly changing as forecasts of future returns and risk are updated.

This research investigates the predictive power of machine learning (ML) architectures (Random Forests, XGBoost, and LSTM networks) against traditional econometric benchmarks (GARCH-family and HAR-RV) for equity volatility forecasting. Using daily and high-frequency data from 2010–2025 across major U.S. indices and fifteen liquid equities, I implement a monthly walk-forward validation framework to compare statistical accuracy (RMSE, QLIKE) with economic utility. While traditional models are the industry standard, they often fail to capture the non-linear regime shifts characteristic of modern markets. This study evaluates whether ML-driven forecasts can generate superior risk-adjusted returns through three portfolio applications: volatility-timing, risk-parity, and a novel regime-based allocation.

The project will culminate in a comprehensive research manuscript and a reproducible Python-based backtesting engine, providing a data-driven assessment of whether the added complexity of machine learning is justified by its economic gains in a post-2020 market environment.

PROPOSAL NARRATIVE

1. Purpose of the Study

The goal of this ten-week research project is to evaluate the predictive efficacy of Machine Learning (ML) architectures, specifically Random Forests, XGBoost, and Long Short-Term Memory (LSTM) networks, against traditional econometric benchmarks like GARCH and HAR-RV. While traditional models are computationally efficient, they often fail to capture the non-linear "regime shifts" and structural breaks seen in modern markets, such as the 2020 COVID-19 crash.

This project explores three central questions:

1. **Statistical Accuracy:** Do ML and hybrid models significantly reduce out-of-sample forecast errors (measured by RMSE and QLIKE) compared to parametric benchmarks?
2. **Economic Utility:** Does superior statistical accuracy translate into higher risk-adjusted returns when applied to volatility-timing and risk-parity portfolio strategies?
3. **Regime Resilience:** How do these models perform during periods of extreme market stress compared to periods of relative calm?

The ultimate goals of this project are:

- To build a robust, reproducible Python-based backtesting engine for volatility forecasting.
- To determine if the added complexity of machine learning is economically justified after accounting for transaction costs.
- To produce a publication-quality manuscript suitable for a journal in financial economics.

2. Rationale

Volatility forecasting is a central problem in financial economics because asset returns exhibit "volatility clustering", periods of high risk followed by gradual reversion. Accurate forecasts are critical for option pricing, Value-at-Risk (VaR) calculations, and portfolio optimization. My interest in this field stems from the tension between classic theory and modern market reality. Engle's (1982) Autoregressive Conditional Heteroskedasticity (ARCH) framework formalized the idea that return variance may depend on past information. Bollerslev (1986) extended this to the Generalized ARCH (GARCH) model, allowing current volatility to depend on both lagged squared returns and lagged volatility. Later extensions address empirically important asymmetries in equity markets. Nelson's (1991) Exponential GARCH (EGARCH) captures leverage effects by allowing negative returns to increase volatility more than positive returns of the same magnitude, while Glosten, Jagannathan, and Runkle's (1993) GJR-GARCH introduces threshold effects to model similar asymmetries.

This research is important to me because it allows me to test the "Hansen and Lunde (2005) paradox", i.e., the finding that the simple GARCH(1,1), where daily volatility depends on yesterday's squared return and yesterday's volatility, remains remarkably difficult to outperform against modern ML architectures. I am fascinated by the possibility that architectures like LSTMs or XGBoost can capture regime-dependent behavior that traditional models miss. However, I am also an economic skeptic; I want to see if these "black box" improvements survive realistic transaction costs.

Most existing literature excludes the critical 2020–2025 period, with the COVID-19 shock and the subsequent high-inflation regime. By using data through 2025, I will provide a fresh assessment of whether ML provides a "free lunch" or if the added complexity is redundant. This project connects deeply to my goal of pursuing quantitative finance, as it requires me to synthesize rigorous econometrics with advanced data science to solve a real-world investment problem.

3. Literature Review

Traditional Volatility Modeling

Engle (1982) introduced ARCH models to account for volatility clustering, a stylized fact inconsistent with constant-variance assumptions. Bollerslev (1986) generalized this framework through GARCH models, which capture persistent volatility dynamics more parsimoniously. Nelson (1991) proposed the EGARCH model to account for leverage effects, while Glosten et al. (1993) developed the GJR-GARCH model to allow asymmetric responses of volatility to positive and negative return shocks. Hansen and Lunde (2005) conduct an extensive comparison of ARCH-type models and conclude that simple GARCH(1,1) specifications are extremely difficult to outperform in forecasting exercises.

A major advance in volatility measurement was introduced by Andersen et al. (2003), who show that realized volatility—constructed by aggregating high-frequency intraday return variances—provides a close empirical proxy for integrated volatility, defined as the cumulative variation of asset prices over a trading day. Although their empirical application focuses on foreign exchange markets, the realized volatility framework has since been widely adopted in equity markets. Building on this insight, Corsi (2009) proposes the Heterogeneous Autoregressive model of Realized Volatility (HAR-RV), which captures volatility persistence across daily, weekly, and monthly horizons and has become a standard benchmark in empirical volatility forecasting.

Machine Learning in Finance

Machine learning methods have gained prominence in financial research due to their ability to model nonlinearities, interactions, and complex dynamics in high-dimensional data. Gu, Kelly, and Xiu (2020) demonstrate that tree-based models and neural networks can outperform traditional linear models in predicting stock returns when processing large sets of firm characteristics. Krauss et al. (2017)

apply deep neural networks, gradient-boosted trees, and Random Forests to statistical arbitrage strategies for S&P 500 stocks, focusing on return predictability rather than volatility forecasting, but emphasizing the critical importance of careful validation to avoid overfitting.

In the specific context of volatility forecasting, Kristjanpoller and Minutolo (2018) find that neural networks outperform GARCH models for stock volatility, while Ryll and Seidens (2019) document the strong performance of Long Short-Term Memory (LSTM) networks in cryptocurrency markets. Bucci (2020) shows that ensemble methods such as Random Forests can improve upon GARCH benchmarks for equity volatility forecasting. However, Liu (2019) provides a more cautious assessment, finding that LSTM models often match but do not substantially exceed GARCH performance, particularly in out-of-sample settings.

Economic Evaluation of Volatility Forecasts

An important strand of the literature emphasizes that improvements in statistical forecast accuracy do not necessarily translate into economic value. West, Edison, and Cho (1993) show that models ranked highly by statistical loss functions may deliver little utility when used for economic decision-making. In other words, a model that makes slightly more accurate numerical predictions may still lead to worse investment decisions if those prediction errors occur at inopportune times. Fleming, Kirby, and Ostdiek (2001, 2003) pioneered the evaluation of volatility forecasts through volatility-timing strategies, demonstrating that improved forecasts can generate economically meaningful gains through dynamic asset allocation. Moreira and Muir (2017) further show that volatility-managed portfolios significantly improve risk-adjusted returns across asset classes.

Methodologically, Hansen and Lunde (2006) introduce Model Confidence Set procedures to account for multiple testing when comparing volatility models, while Patton (2011) shows that volatility model rankings can remain robust even when using imperfect volatility proxies.

Research Positioning

This study contributes to the literature by providing a comprehensive and up-to-date comparison of traditional econometric and machine learning volatility models using daily equity data through 2025. It evaluates multiple machine learning architectures under rigorous walk-forward validation, combines statistical and portfolio-based economic evaluation, and leverages high-quality data sources to assess whether forecast improvements translate into real economic value.

4. Research Design

This research employs a structured empirical design that integrates traditional econometric models, modern machine learning techniques, and portfolio-based economic evaluation to assess the practical value of volatility forecasting innovations.

Data Sources and Sample Construction

The study uses daily data from January 2010 through December 2025, yielding approximately 4,000 observations per asset. Financial data are obtained primarily from the Bloomberg Terminal and supplemented with publicly available sources such as the CBOE and FRED. The sample includes three major U.S. equity exchange-traded funds—SPDR S&P 500 ETF Trust (SPY), Invesco QQQ Trust (QQQ), and iShares Russell 2000 ETF (IWM)—as well as fifteen highly liquid U.S. stocks across multiple sectors: technology (Apple, Microsoft, NVIDIA), financials (JPMorgan Chase, Goldman Sachs, Bank of America), healthcare (Johnson & Johnson, UnitedHealth Group, Pfizer), consumer (Amazon, Walmart, Home Depot), and energy/industrials (Exxon Mobil, Chevron, Caterpillar). This diversified cross-section reduces the likelihood that results are driven by sector-specific dynamics.

Forecasting Models

The analysis compares eight volatility forecasting models spanning three categories: traditional econometric specifications, machine learning approaches, and hybrid methods.

Traditional econometric models include GARCH(1,1) as the baseline. The GARCH(1,1) model specifies that today's conditional variance depends on yesterday's squared return (recent market shock) and yesterday's variance (persistence). Mathematically:

$$\sigma^2_t = \omega + \alpha \cdot r^2_{t-1} + \beta \cdot \sigma^2_{t-1}$$

where σ^2_t represents forecasted variance for day t, r^2_{t-1} is yesterday's squared return, and ω, α, β are estimated parameters. In plain terms, if markets were turbulent yesterday (large r^2_{t-1}), the model predicts elevated volatility today. High persistence occurs when $(\alpha + \beta) \approx 1$, meaning volatility shocks decay slowly—a common finding in equity markets.

EGARCH(1,1) captures the empirical fact that negative returns increase volatility more than positive returns of equal magnitude—the leverage effect. This asymmetry is modeled as:

$$\log(\sigma^2) = \omega + \alpha \cdot |z_{-1}| + \gamma \cdot z_{-1} + \beta \cdot \log(\sigma^2_{-1})$$

where $z_{-1} = r_{-1}/\sigma_{-1}$ is the standardized return. The asymmetry parameter γ captures leverage: when $\gamma < 0$, negative shocks ($z_{-1} < 0$) produce larger volatility increases. The logarithmic specification ensures σ^2 remains positive without parameter restrictions.

GJR-GARCH(1,1) provides an alternative threshold specification for modeling asymmetry through a piecewise linear structure, offering parsimony while capturing the leverage effect that characterizes equity volatility.

HAR-RV (Heterogeneous Autoregressive Realized Volatility) recognizes that volatility exhibits persistence across multiple time horizons—daily, weekly, and monthly. The model is specified as:

$$RV_t = \beta_0 + \beta^d \cdot RV_{t-1} + \beta^w \cdot RV_{t-5:t-1} + \beta^m \cdot RV_{t-22:t-1} + \varepsilon$$

where RV_t is today's realized volatility, RV_{t-1} is yesterday's (daily component), $RV_{t-5:t-1}$ is the average over the past 5 days (weekly), and $RV_{t-22:t-1}$ is the average over the past 22 days (monthly).

Machine learning approaches include Random Forest with 500-1,000 decision trees trained using bootstrap aggregating, capturing complex nonlinear relationships while reducing overfitting through averaging. XGBoost implements sequential boosting where each new tree fits residuals from the previous ensemble, with early stopping preventing overfitting by terminating training when validation loss ceases improving. Long Short-Term Memory (LSTM) neural networks utilize recurrent connections designed for sequential data, processing 20-60 day lookback windows through 2-3 LSTM layers (64-128 units each) with dropout regularization (rates 0.2-0.3). All machine learning models undergo rigorous hyperparameter tuning via time-series cross-validation on the training period.

The hybrid approach combines GARCH(1,1) fitted volatilities and standardized residuals with Random Forest or XGBoost, testing whether machine learning adds incremental predictive power beyond parametric structure. This two-stage procedure allows flexible algorithms to capture residual patterns that GARCH's fixed functional form cannot accommodate.

Feature Engineering

Machine learning models use approximately 40–50 predictors grouped into five categories. Historical volatility features include lagged realized volatility (1, 2, 3, 5, 10, 20, and 60 days), rolling volatility averages (5, 20, and 60 days), and volatility momentum measures. Return-based features include lagged and absolute returns, rolling skewness and kurtosis, and extreme-value indicators over 20-day windows. Market microstructure features include high–low price ranges, volume scaled by recent averages, gap frequencies, and bid–ask spread proxies where available. Market-wide information incorporates VIX levels and changes, S&P 500 indicators for individual-stock models, and market breadth measures. Calendar effects capture day-of-week, month-of-year, and post–Federal Reserve announcement patterns.

Highly redundant predictors are removed using correlation filtering (absolute pairwise correlation exceeding 0.95). Additional feature selection is performed using Random Forest importance rankings and LASSO (least absolute shrinkage and selection operator) regularization when necessary to mitigate overfitting and reduce computational burden.

Walk-Forward Validation

Model performance is evaluated using expanding-window walk-forward validation with monthly refitting. Models are initially trained using data through December 2017, generate forecasts for January 2018, and are subsequently refit each month as new data becomes available through December 2025. At each refit, one-day, five-day, and twenty-day ahead volatility forecasts are produced using only information available at the forecast date. This procedure ensures strict out-of-sample evaluation and avoids look-ahead bias, closely mirroring real-time forecasting conditions faced by market participants.

Statistical Evaluation

Forecast accuracy is evaluated using multiple complementary loss functions. Root Mean Squared Error (RMSE) penalizes large forecast errors and is sensitive to tail events relevant for risk management, while Mean Absolute Error (MAE) provides a robust measure less affected by outliers. Volatility-specific performance is assessed using the Quasi-Likelihood (QLIKE) loss function:

$$QLIKE = \frac{1}{T}\sum_{t=1}^{T}\left(\frac{RV_t}{(\widehat{\sigma}_t)^2} - log\left(\frac{RV_t}{(\widehat{\sigma}_t)^2}\right) - 1\right),$$

where RV_t denotes realized volatility and $(\widehat{\sigma}_t)^2$ is the forecasted variance. QLIKE remains consistent under imperfect volatility proxies. Directional accuracy measures whether models correctly predict the sign of volatility changes.

Statistical significance is assessed using the Model Confidence Set (MCS) procedure of Hansen and Lunde (2006), which accounts for multiple model comparisons and identifies sets of models statistically indistinguishable from the best performer at 90% and 95% confidence levels.

Economic Evaluation

Economic evaluation tests whether forecast improvements translate into better investment outcomes under realistic trading conditions. Portfolio strategies are assessed using Sharpe ratios, Sortino ratios, maximum drawdowns, Calmar ratios, and transaction cost drag.

Volatility-Timing Strategy

A volatility-timing strategy dynamically allocates between SPY and SHY to target constant risk:

$$w_t^{SPY} = min(1.5,\ max(0, \frac{\sigma^*_{target}}{\widehat{\sigma}_t})),$$

where σ^*_{target}=15% and $\widehat{\sigma}_t$ is the forecasted SPY volatility. The strategy rebalances monthly, prohibits short positions, caps leverage at 150%, and applies $5 per-trade transaction costs. Benchmarks include buy-and-hold SPY and static 60/40 portfolios.

Cross-Sectional Risk-Parity Strategy

A risk-parity portfolio of fifteen stocks assigns weights inversely proportional to forecasted volatility,

$$w_i = \frac{1/\hat{\sigma}_i}{\Sigma_j 1/\hat{\sigma}_j},$$

subject to position and turnover constraints. Performance is compared to equal-weighted and market-cap-weighted benchmarks.

Regime-Based Allocation

A regime-based strategy adjusts equity–bond exposure discretely across low (<15%), medium (15–25%), and high (>25%) volatility regimes, trading less frequently than continuous targeting. Regime frequencies are analyzed to assess whether model performance is driven by common or stress environments.

Across all strategies, performance attribution decomposes returns into forecast accuracy, market timing, alpha, and transaction cost effects.

5. Work Plan

Period	Focus	Key Tasks & Deliverables
Week 1	Literature Review & Data Collection	• Comprehensive literature review • Extract securities data, VIX, and other market condition data
Week 2	Exploratory Analysis & Baseline	• Exploratory data analysis (volatility clustering, distributions, stylized facts) • Compute realized volatility from intraday data • Implement GARCH(1,1) baseline • **Advisor meeting:** review preliminary findings
Week 3	Traditional Econometric Models	• Implement EGARCH, GJR-GARCH, HAR-RV • MLE estimation and diagnostics • Walk-forward out-of-sample forecasts • **Advisor meeting:** discuss estimation results
Week 4	Feature Engineering & Machine Learning Setup	• Construct machine learning feature set • Handle missing data and feature scaling • Create train/validation/test splits • Correlation analysis to remove redundancy • **Advisor meeting:** validate feature engineering
Week 5	Random Forest & XGBoost	• Random Forest with time-series CV and tuning • XGBoost with early stopping • Generate out-of-sample forecasts and analyze feature importance • **Advisor meeting:** review machine learning results
Week 6	LSTM & Hybrid Models	• Design and implement an LSTM architecture • Train with dropout and early stopping

		• Develop a GARCH-Machine Learning hybrid model • Generate forecasts from all models • **Advisor meeting:** discuss NN performance
Week 7	Statistical Evaluation	• Compute RMSE, MAE, QLIKE, and model Confidence Set tests • Analyze performance by volatility regime and by crisis periods • **Advisor meeting:** interpret statistical results
Week 8	Economic Evaluation	• Volatility-timing strategies • Risk-parity portfolio backtests • Regime-based tactical allocation • Compute Sharpe ratios, drawdowns, transaction costs • **Advisor meeting:** validate portfolio methodology
Week 9	Robustness & Extensions	• Tests across stocks and sectors • Sensitivity analysis (costs, constraints) • Performance attribution • Cost-benefit synthesis • **Advisor meeting:** discuss implications
Week 10	Documentation & Presentation	• Publication-quality visualizations • Draft full research paper • Prepare Fall Symposium poster • Final project reflection • **Final advisor meeting:** review and publication strategy
Post-Program	Dissemination & Applications	• Revise paper based on feedback • Submit to undergraduate journals • Present at Fall Research Symposium • Integrate into grad school applications

Appendix

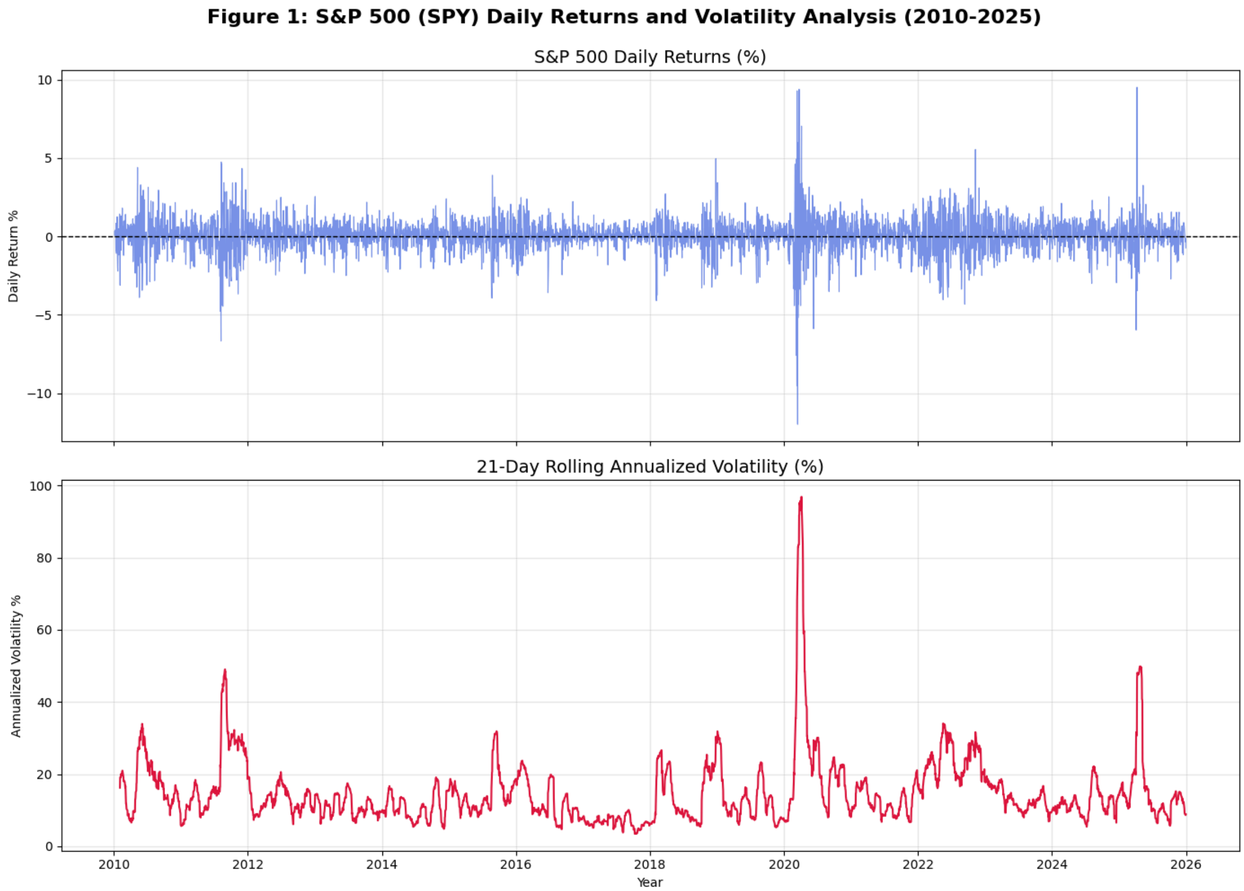

Figure 1 illustrates the time‑varying nature of equity market volatility using daily S&P 500 returns from 2010–2025. Volatility is measured as the annualized 21‑day rolling standard deviation of returns. Periods such as the European debt crisis, the COVID‑19 market collapse, and the post‑pandemic tightening cycle exhibit sharp volatility spikes followed by gradual mean reversion. This behavior motivates the use of dynamic volatility models rather than constant‑variance assumptions.

Figure 2: Volatility Clustering via Squared Returns (SPY, 2010-2025)

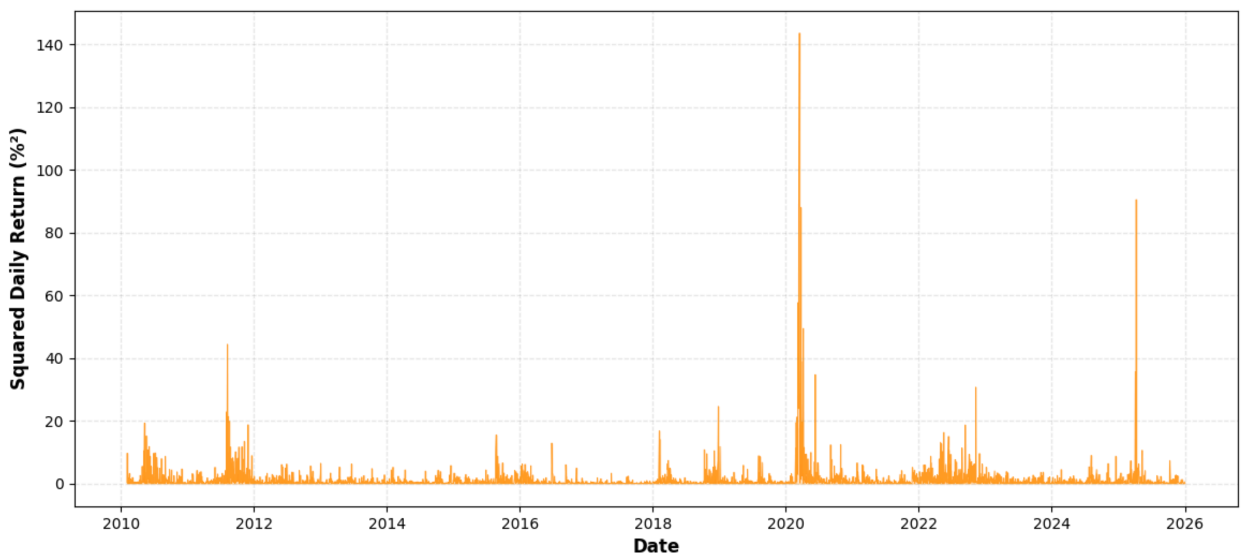

Figure 2 presents daily squared returns of the S&P 500 ETF (SPY), 2010–2025. Large return shocks tend to cluster in time rather than occurring independently, providing visual evidence against constant-variance assumptions and motivating conditional volatility models such as GARCH.

Figure 3: Volatility Persistence - Autocorrelation of Realized Volatility (SPY, 2010-2025)

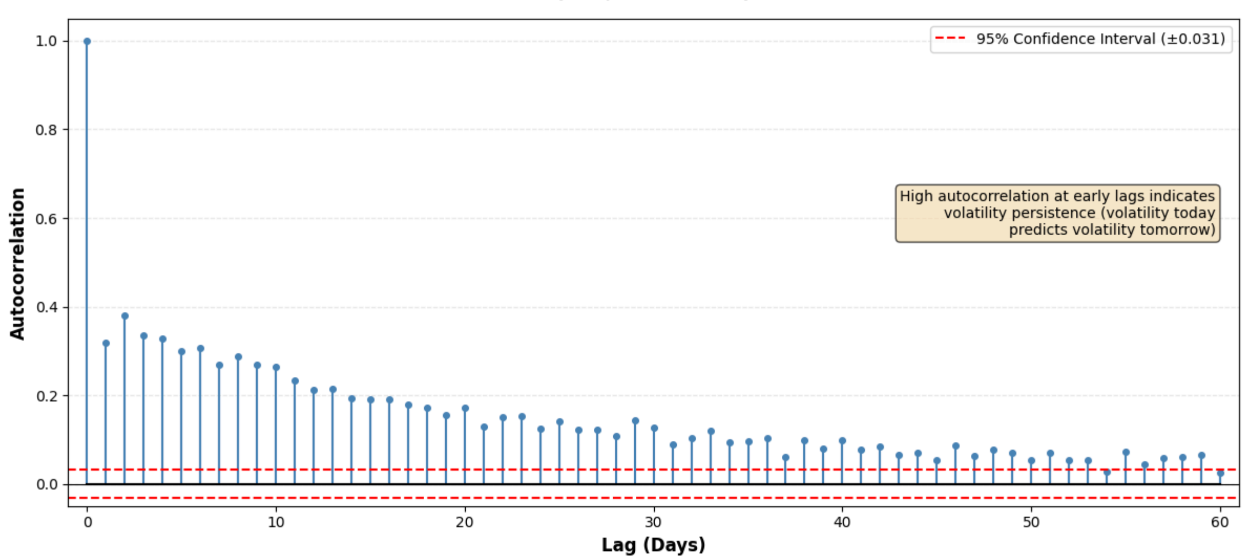

Figure 3 shows the autocorrelation of daily realized volatility for SPY. Volatility exhibits strong persistence over multiple horizons, supporting the use of dynamic forecasting models rather than static variance estimates.

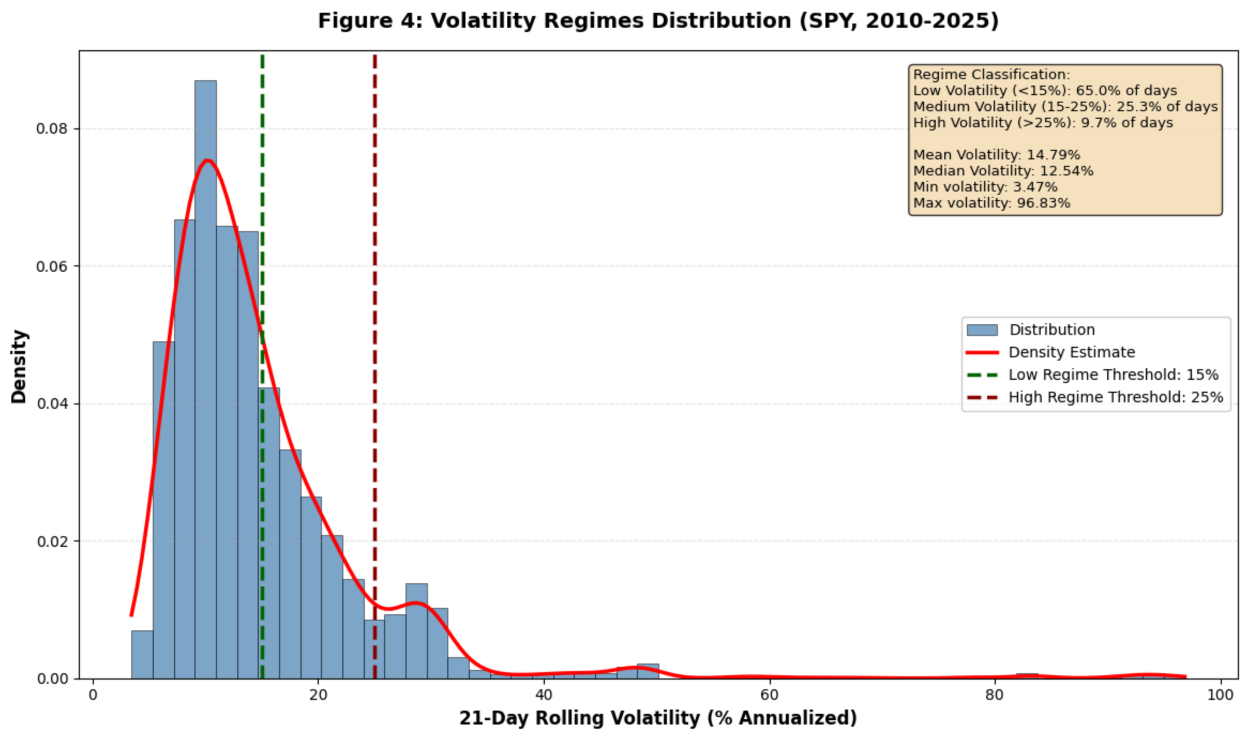

Figure 4 demonstrates the distribution of 21-day rolling annualized volatility for SPY. The distribution exhibits clear separation between low-, medium-, and high-volatility regimes, motivating regime-based evaluation of forecasting models and portfolio strategies.

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